News Bias Detector

Zero Shot Zealots
Albert, Prathit, Sandesh

Problem

HEADLINE ROUNDUP

Federal Reserve Holds Interest Rates Steady Amid Economic Uncertainty

From the Left

Fed holds interest rates steady, defying pressure from Trump

ABC News (Online) L L C R R

From the Center

Fed Maintains Interest Rates But Warns Of 'Uncertainty' Amid Tariff Turmoil

Newsweek L L C R R

From the Right

Federal Reserve holds key interest rate steady amid economic uncertainty

Fox Business L L C R R

- Detect bias of news articles based on headlines and content
- Multi-class categorization: identifying if an article leans left, center, or right
- Experiment with different architectures and deep learning techniques
- Train on scraped data from AllSides
  - US-based news source that presents “all sides” of political events/topics/opinions
- Previous work on news bias suggests deep learning techniques can detect political bias – particularly using the BERT pre-trained models

Dataset

Qbias Dataset from IR Group

- GitHub Repo released 2023 with:
  - Pre-collected data from AllSides
- All based on All-Sides labels from professionals
  - Each article labeled as left, right, or center
- Dataset creation paper: Qbias - A Dataset on Media Bias in Search Queries and Query Suggestions
- Data Prep: lower case, data cleansing, concatenating into one single column of text for training

Method

1. Use the title, heading, source, tags, and main article text for training

2. Tokenize using the BERT tokenizer and used it to train a custom LSTM as our baseline model

4. Next, train a term frequency-inverse document frequency (TF-IDF) – a common NLP modeling method

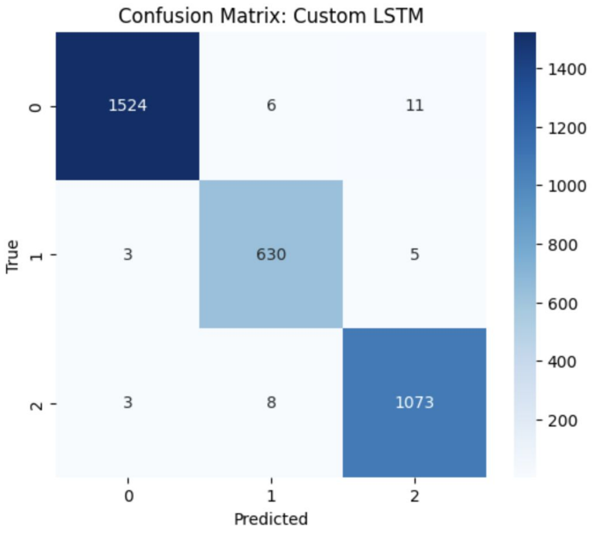
5. Fine-tune BERT (medium) as a more complex model

6. Evaluate all models

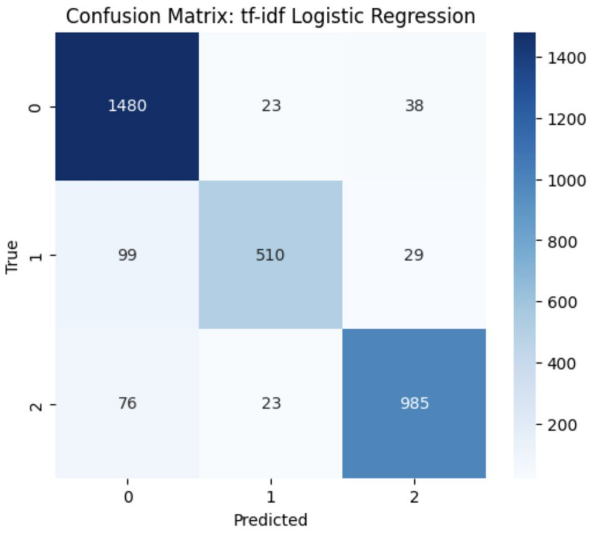
Evaluation Techniques

- For each of our models, we created a confusion matrix with a heatmap to understand the correct/incorrect classifications
- We also examined some good/bad predictions to understand why samples were being classified incorrectly based on their confidence scores
- Analyze ROC curves to analyze performance and softmax distributions to understand confidence

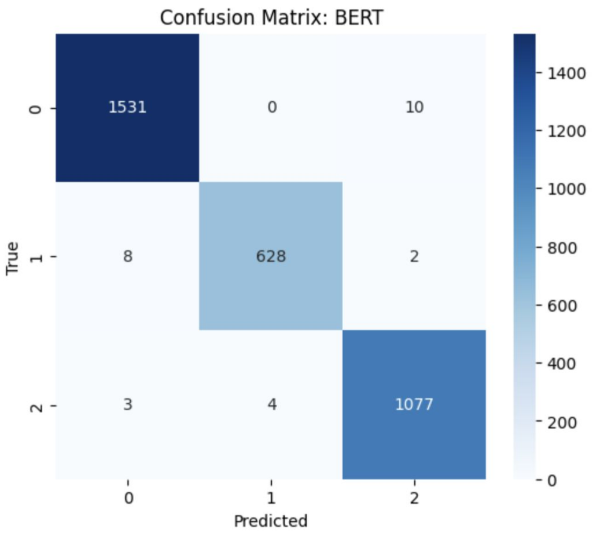
Key Results



Confusion Matrix: Custom LSTM
1524
6
11
3
630
5
3
8
1073
True
Predicted
0
1
2
200
400
600
800
1000
1200
1400



Confusion Matrix: tf-idf Logistic Regression
1480
23
38
99
510
29
76
23
985
True
Predicted
0
1
2
200
400
600
800
1000
1200
1400

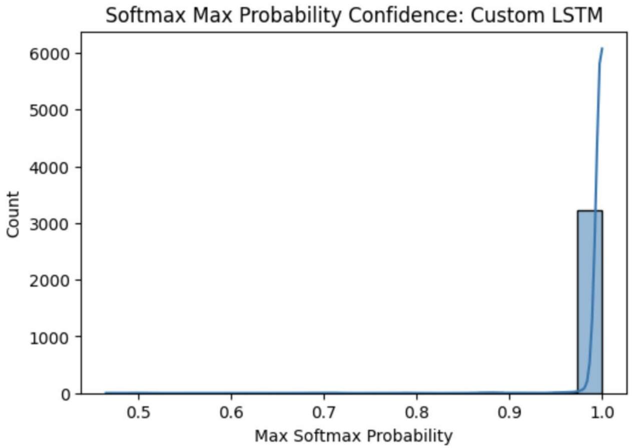


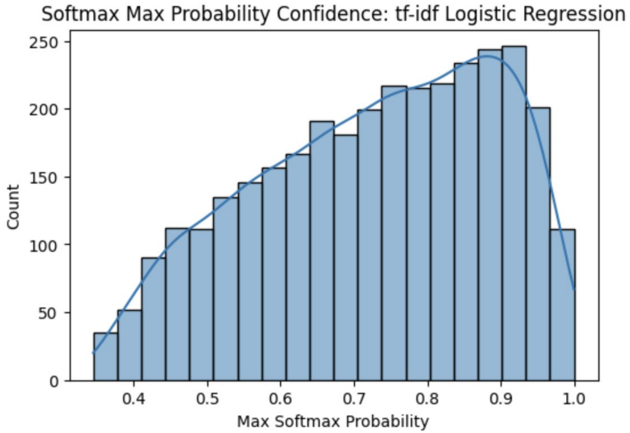
Confusion Matrix: BERT
1531
0
10
8
628
2
3
4
1077
True
Predicted
0
1
2
200
400
600
800
1000
1200
1400

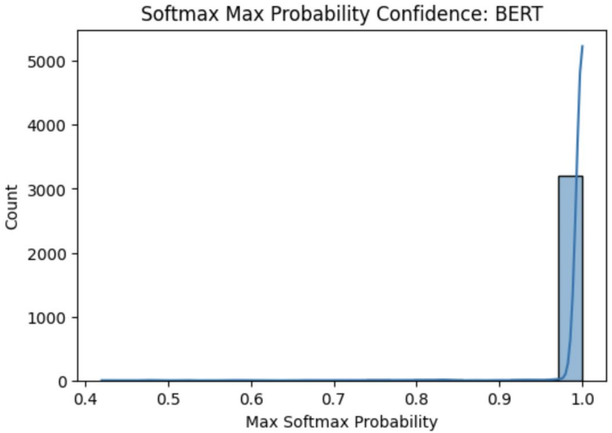
Key Results

Accuracy Scores
→ **LSTM:** 0.989, **TF-IDF:** 0.912, **BERT Fine-tuned:** 0.992

AUC Scores
→ **LSTM:** (1.0, 1.0, 1.0), **TF-IDF:** (0.98, 0.97, 0.98), **BERT Fine-tuned:** (1.0, 1.0, 1.0)







Key Results

Sarah Sanders becomes the latest ex-Trump official to join Fox News

By Oliver Darcy, CNN Business

2 minute read · Updated 4:32 PM EDT, Thu August 22, 2019

Conclusions

- News source is a significant indicator for bias
- Despite discrepancies in data distribution, all models performed quite well
- Our custom LSTM performed just as well as the more advanced fine-tuned BERT model using far less compute power
- Confidently incorrect predictions mainly stemmed from politically charged keywords that the model classified as being right or left leaning

Potential Applications: be more aware of your media consumption/biases, be a more conscious political writer